

P1

- Likelihood function: forget the domain of x , -1pts
- Sufficient statistics: use sample mean instead of sample product, -2pts
- Consistency of $\hat{\theta}$: try to prove it by definition, instead of utilizing the consistency of $\bar{x}$, -3pts
- Reason for $\bar{x}$ to be consistent, -1pts

P2

- Abusing notation for order statistics, -1pts
- Q4, reason for estimator and consistency, -1/2pts
- Q4, MLE is NOT always consistent. -1pts *This is important.*

P3

- Q2: unless give correct rejection region, otherwise for all people who use χ^2 distribution, -3pts *Actually, it's strange to use χ^2 .*
- Q2: use CLT to give an approximate test, -3pts
- Q3: The distribution under H_1 is known, if didn't specify the distribution of $\hat{\beta}$ under H_1 , -1pts
- Q2: Although I took only 3pts for people who use $\left| \frac{2n\alpha\hat{\beta}}{\beta_0} \right| > \chi_{2n\alpha, \omega/2}^2$, it's a severe mistake, since it violates the

intuition of hypothesis testing, and χ^2 is always non-negative.

P4

- Didn't check assumptions, -3pts
- Use t-test, -5pts
- Wrong conclusion or critical value, -1/2pts
- Wrong formula for p-value, -2pts
- Didn't involve cdf function, -1pts

Some interesting "typos" when I was grading:

MLE $\rightarrow$ MME/MLL

Null hypothesis $\rightarrow$ None hypothesis

Skewed $\rightarrow$ showed